Data Structures and Algorithms

LECTURE 05: SORTING ALGORITHMS

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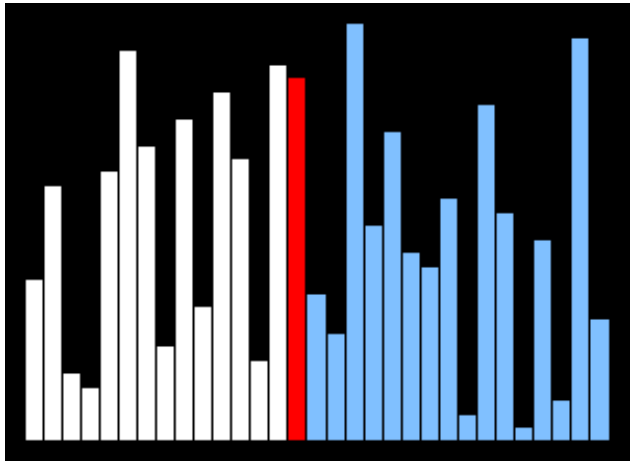
Alliance with FPT Education

Contents

- Simple Sorting Algorithms
  - Selection and Bubble Sort
- Shuffling
- Advanced Sorting Algorithms
  - QuickSort, MergeSort, BucketSort

What is a Sorting Algorithm?

- **Sorting algorithm**
  - An algorithm that rearranges elements in a list
    - In non-decreasing order
  - Elements must be **comparable**
- More formally
  - The **input** is a sequence / list of elements
  - The **output** is an rearrangement / **permutation** of elements
    - In non-decreasing order



Sorting – Example

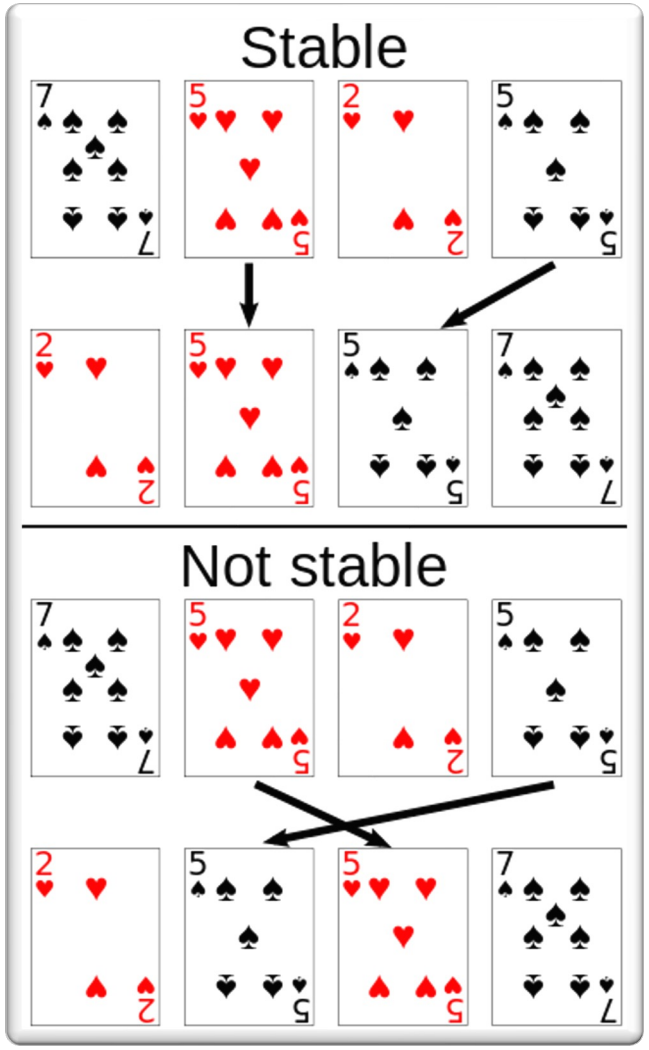
- Efficient sorting algorithms are important for:
  - Producing human-readable output
  - Canonicalizing data – making data uniquely arranged
  - In conjunction with other algorithms, like binary searching
- Example of sorting:

Sorting Algorithms: Classification

- Sorting algorithms are often classified by:
  - Computational **complexity** and memory usage
    - Worst, average and best case behavior
  - **Recursive** / non-recursive
  - **Stability** – stable / unstable
  - **Comparison-based** sort / non-comparison based
  - Sorting **method**: insertion, exchange (bubble sort and quicksort), selection (heapsort), merging, serial / parallel, etc.

Stability of Sorting

- **Stable** sorting algorithms
  - Maintain the order of equal elements
  - If two items compare as equal, their relative order is preserved
- **Unstable** sorting algorithms
  - Rearrange the equal elements in unpredictable order
- Often **different elements** have **same key** used for equality comparing



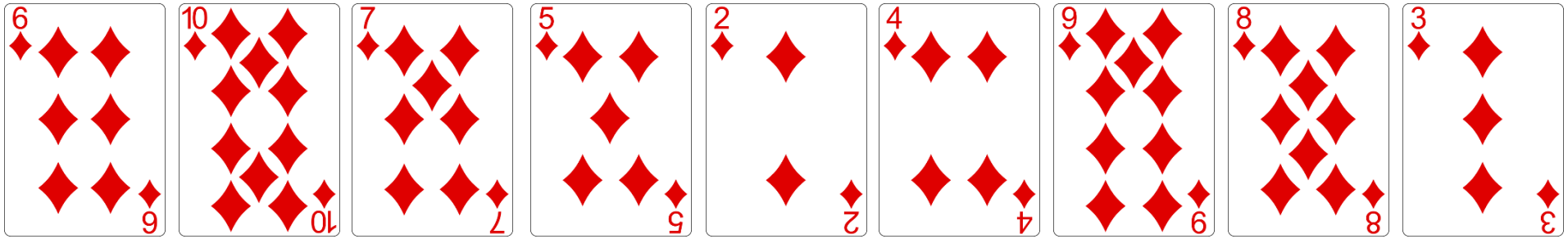
Selection Sort

- **Selection sort** – simple, but inefficient algorithm
  - Swap the first with the min element on the right, then the second, etc.
  - Memory: O(1)
  - Stable: No
  - Method: Selection

Selection Sort Visualization

Index

Min



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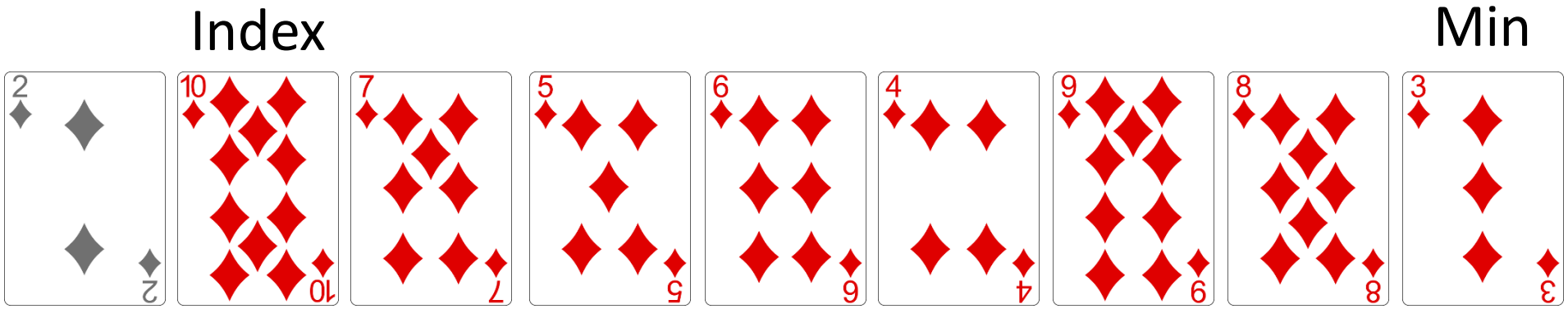

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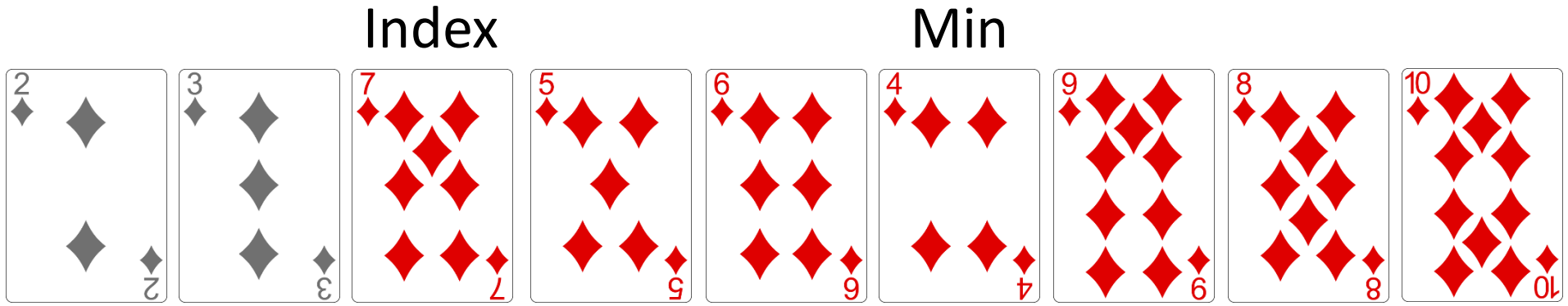

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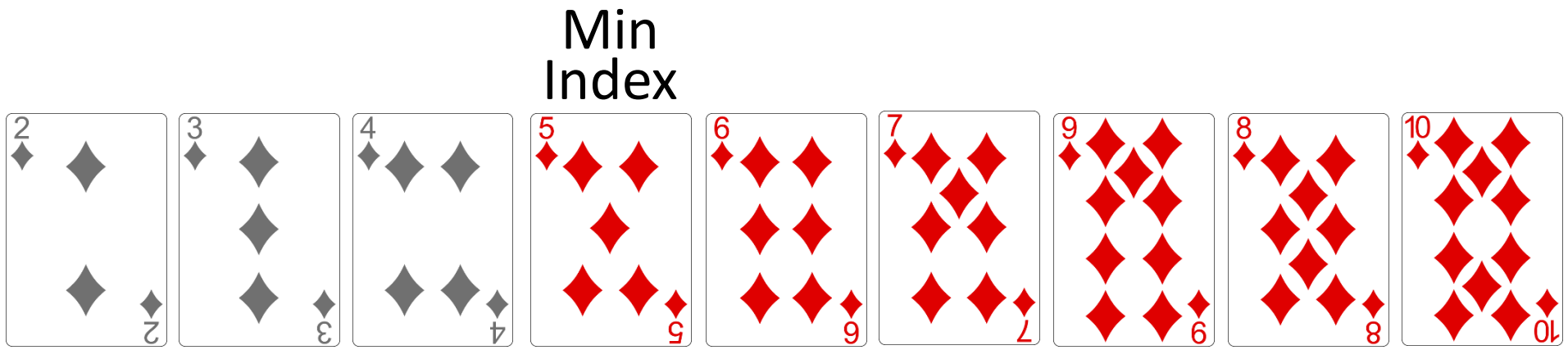

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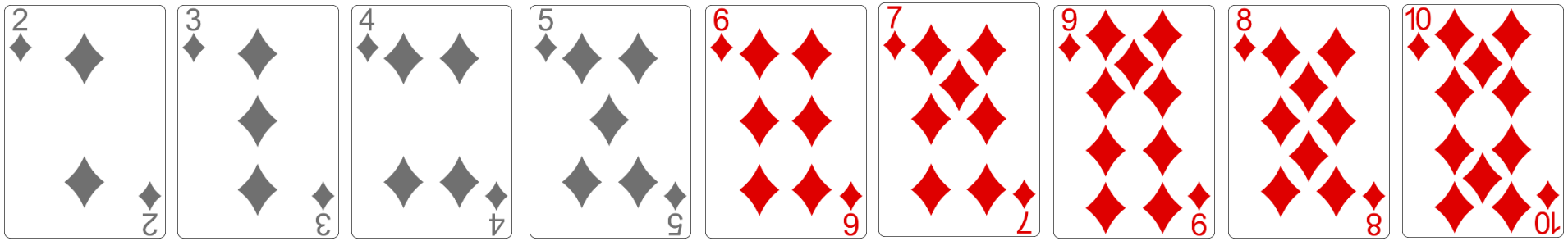

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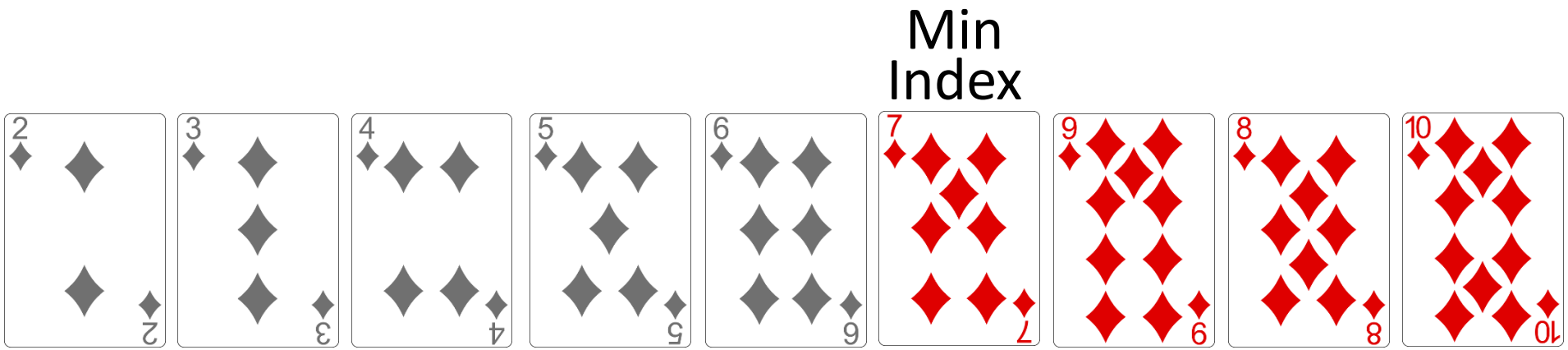

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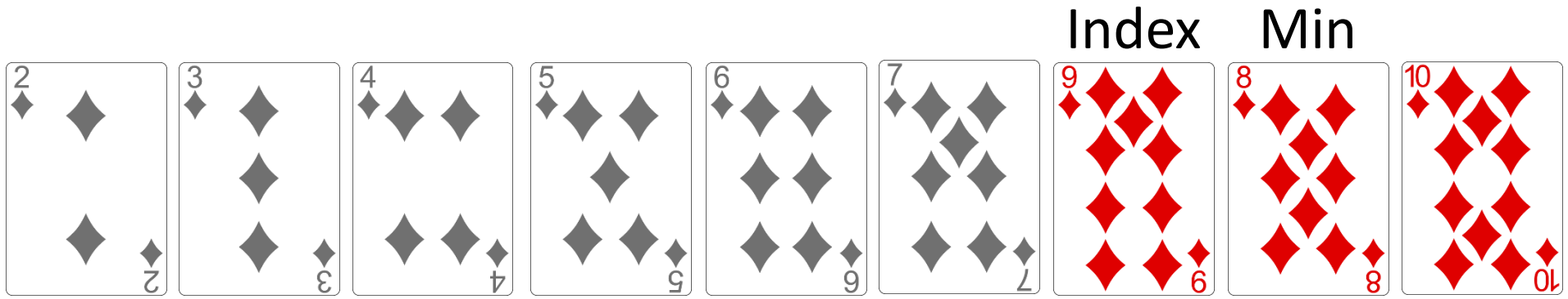

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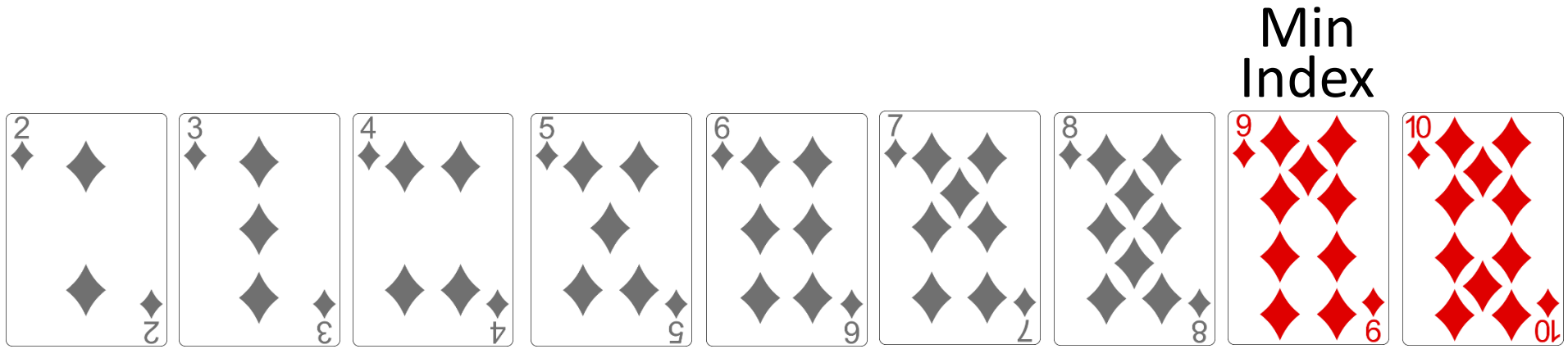

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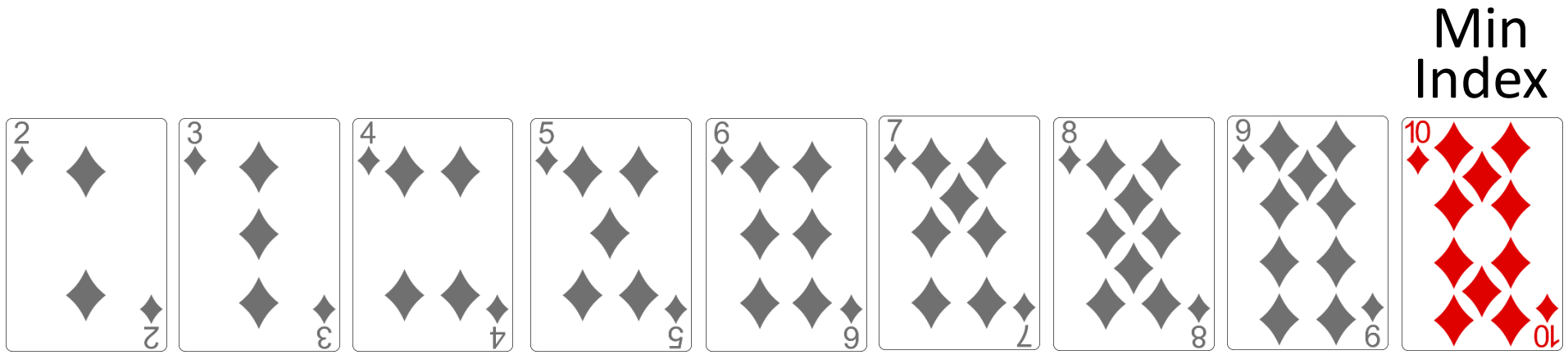

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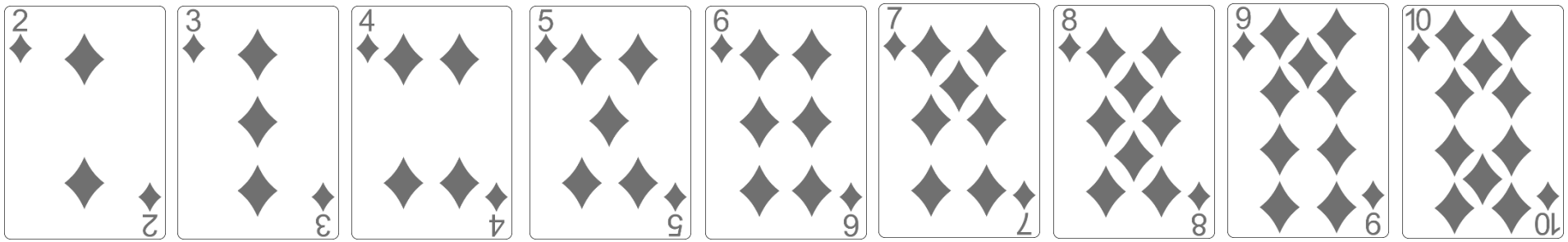

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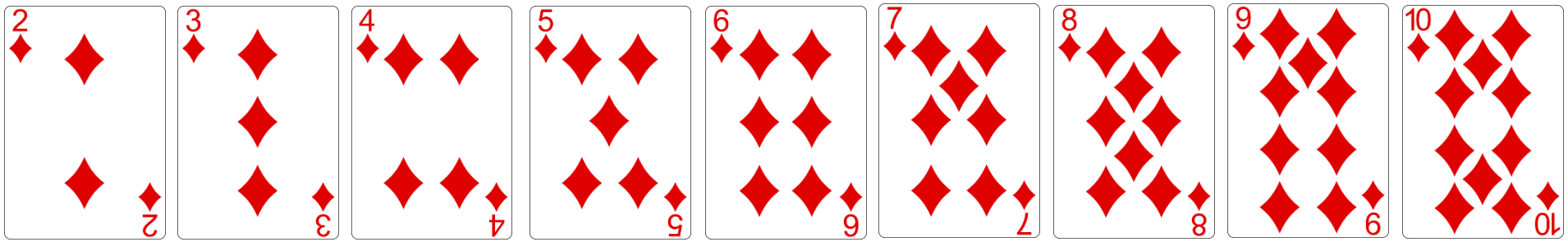


Selection Sort Code

```
for (int index = 0; index < arr.length; index++) {
    int min = index;
    for (int curr = index + 1; curr < arr.length; curr++) {
        if (arr[curr] < arr[min]) {
            min = curr;
        }
    }
    swap(arr, index, min);
}
```

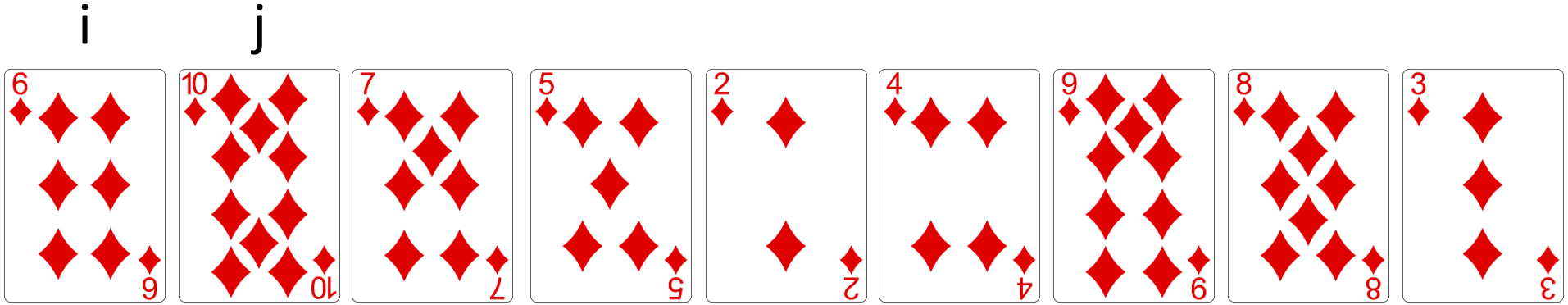
Comparison of Sorting Algorithms

Name	Best	Average	Worst	Memory	Stable	Method
Selection	n^2	n^2	n^2	1	No	Selection

Bubble Sort

- **Bubble sort** – simple, but inefficient algorithm
- Swaps to neighbor elements when not in order until sorted
  - Memory: **O(1)**
  - Stable: Yes
  - Method: Exchanging

Bubble Sort Visualization



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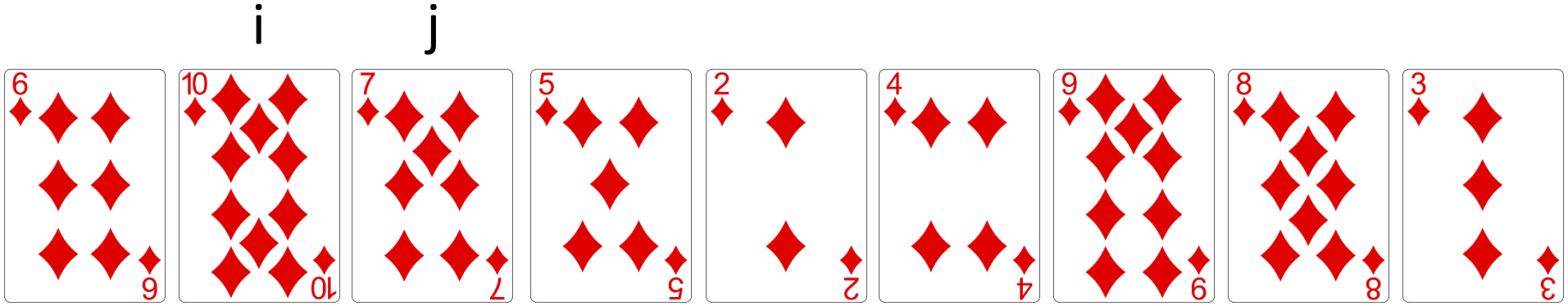

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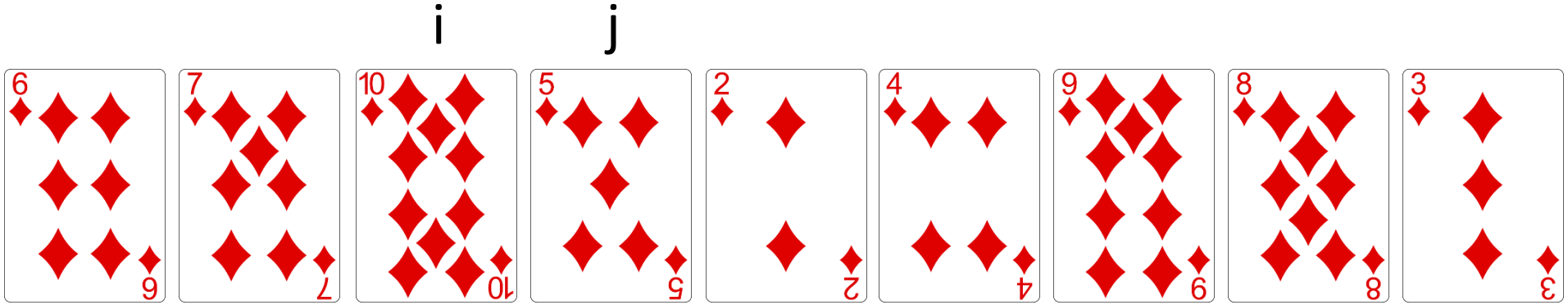

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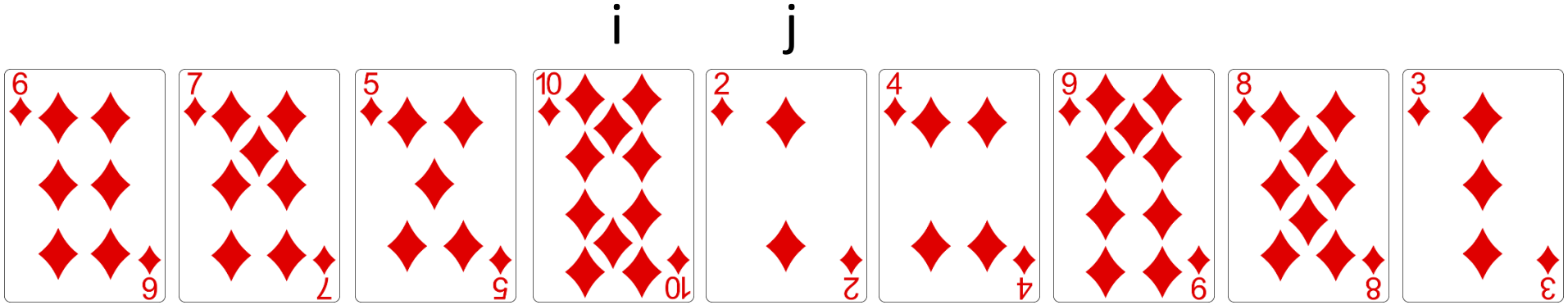

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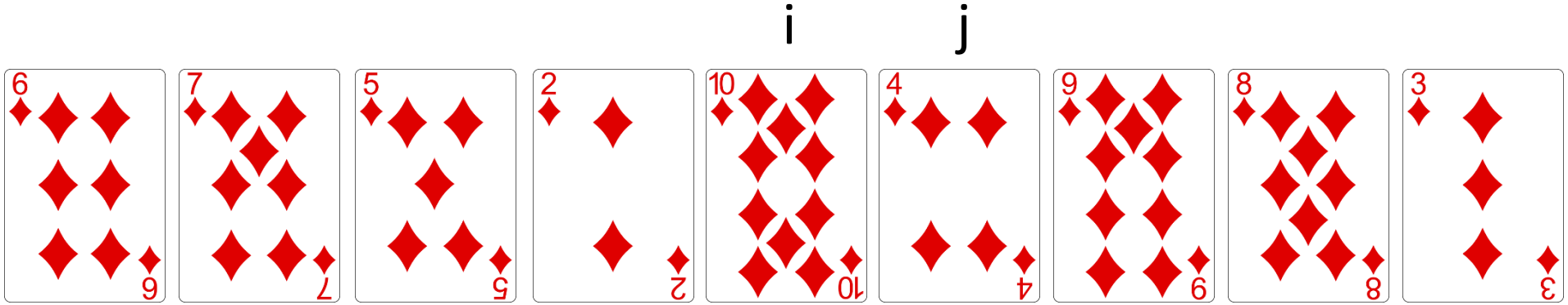

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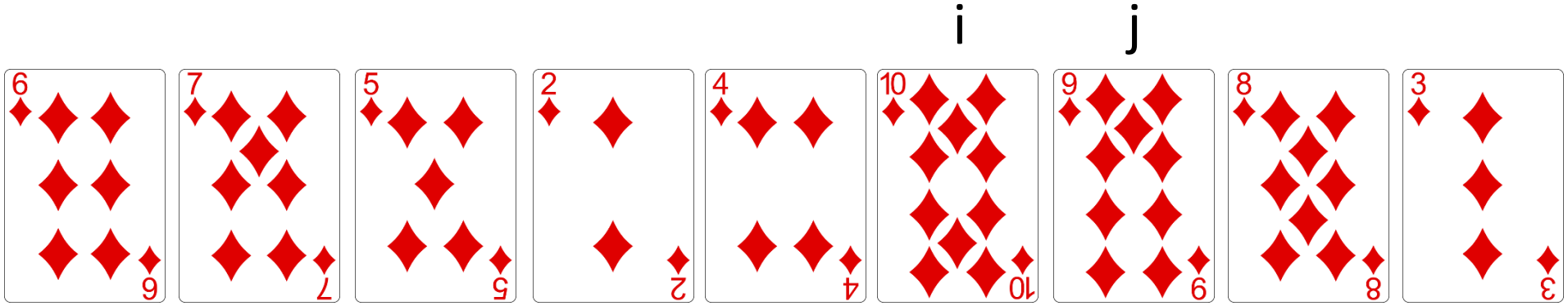

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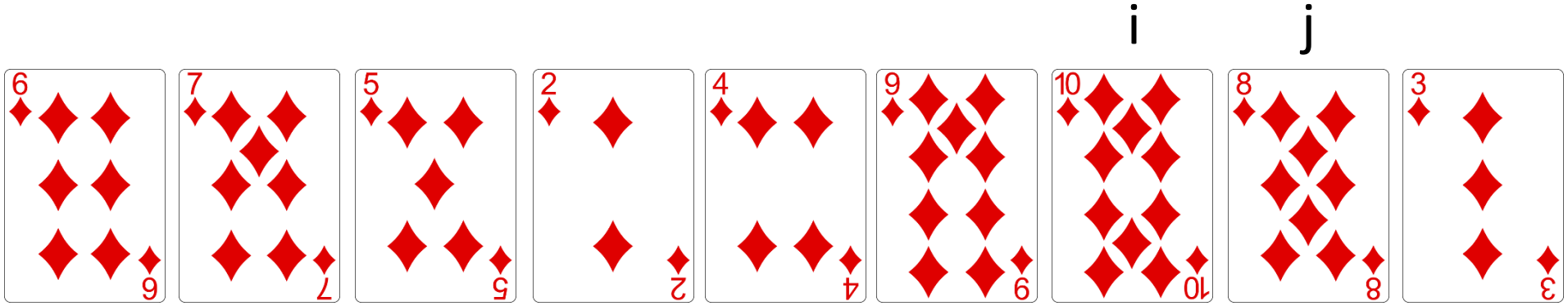

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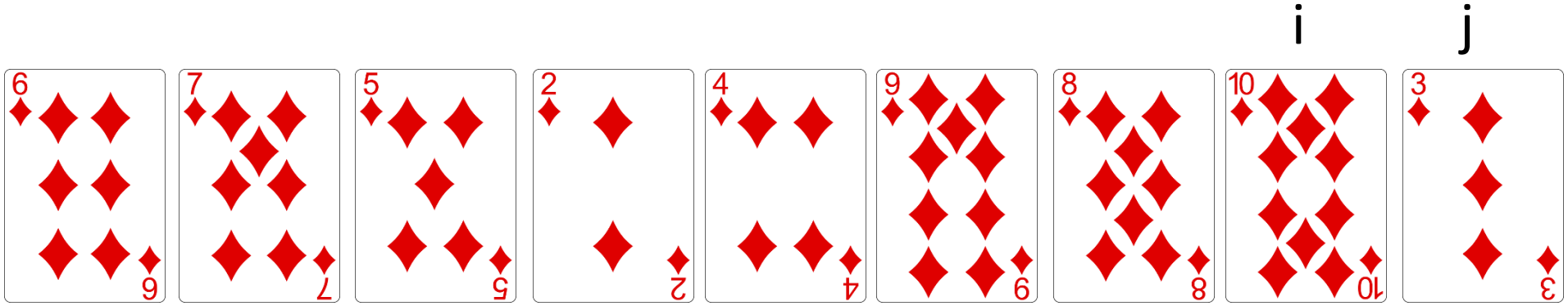

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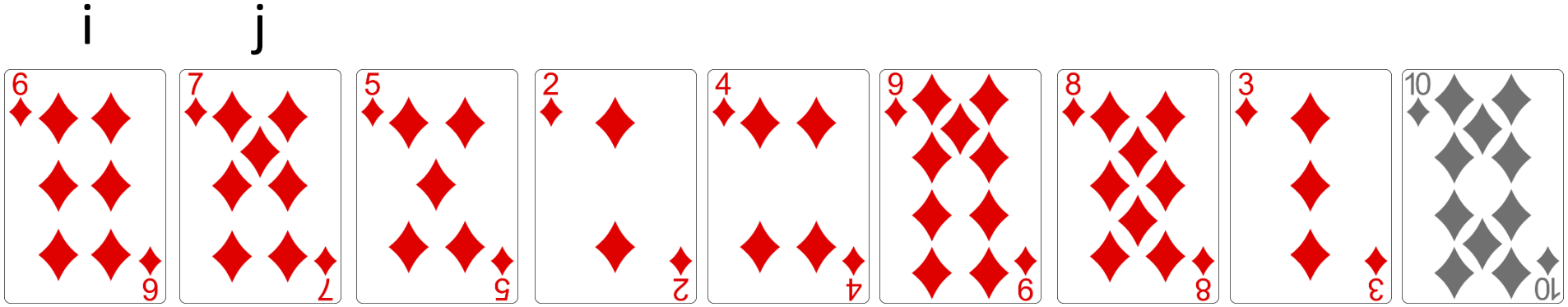

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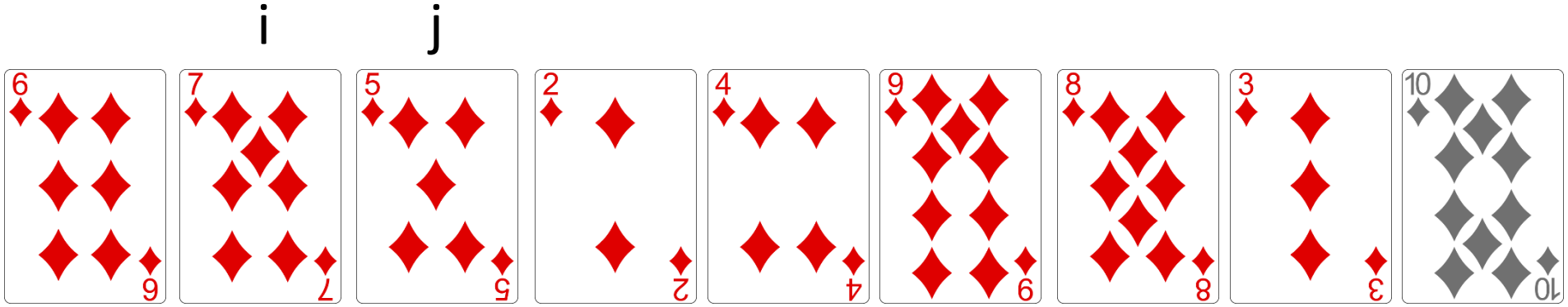

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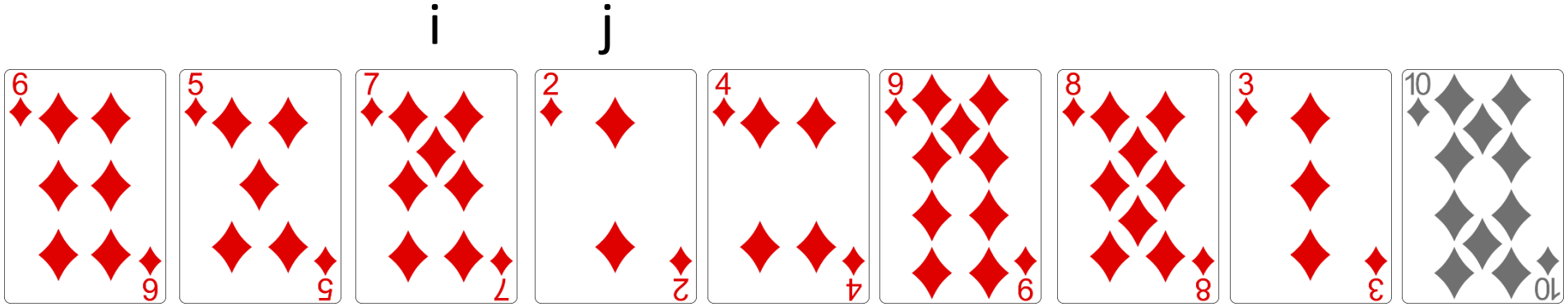

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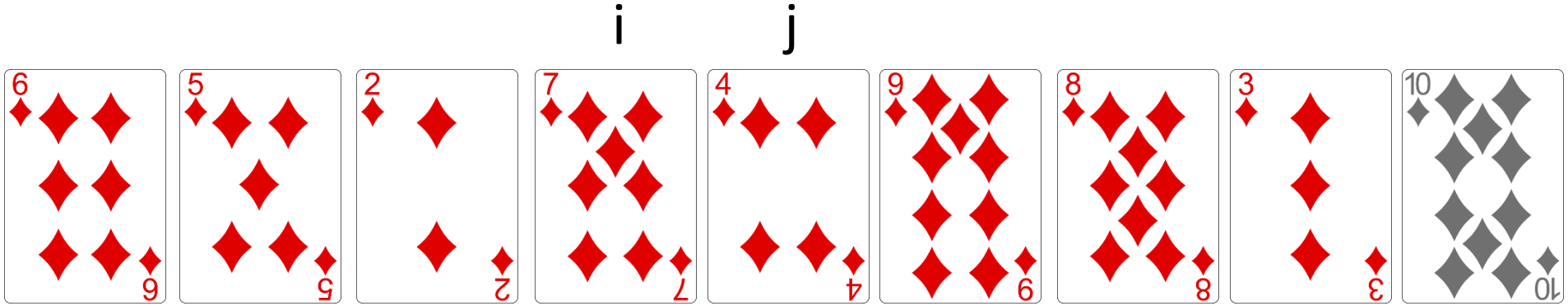

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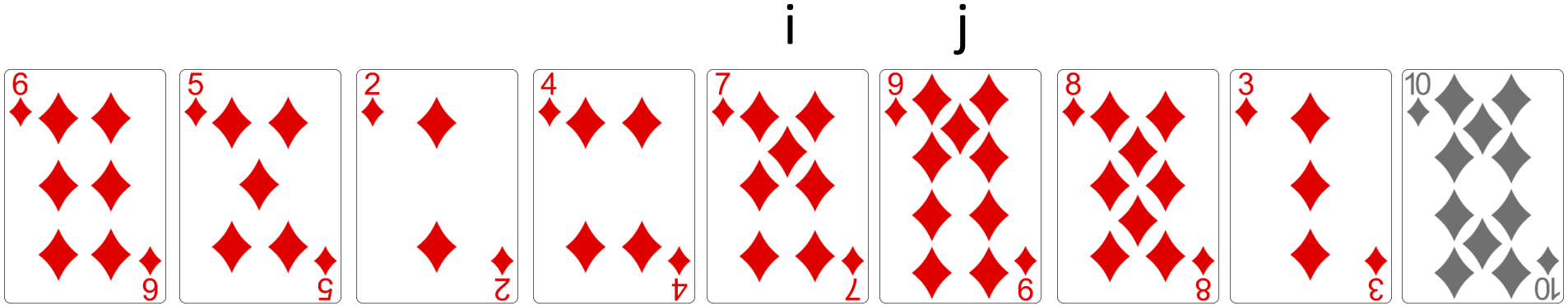

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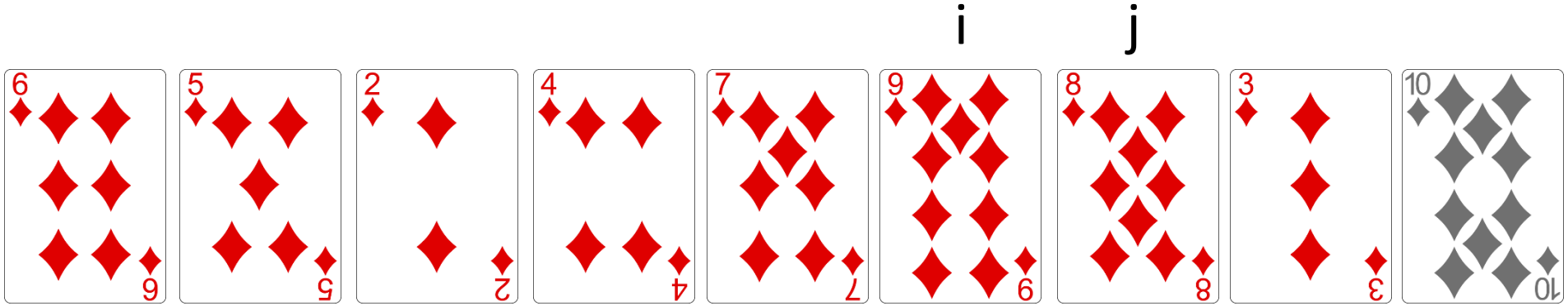

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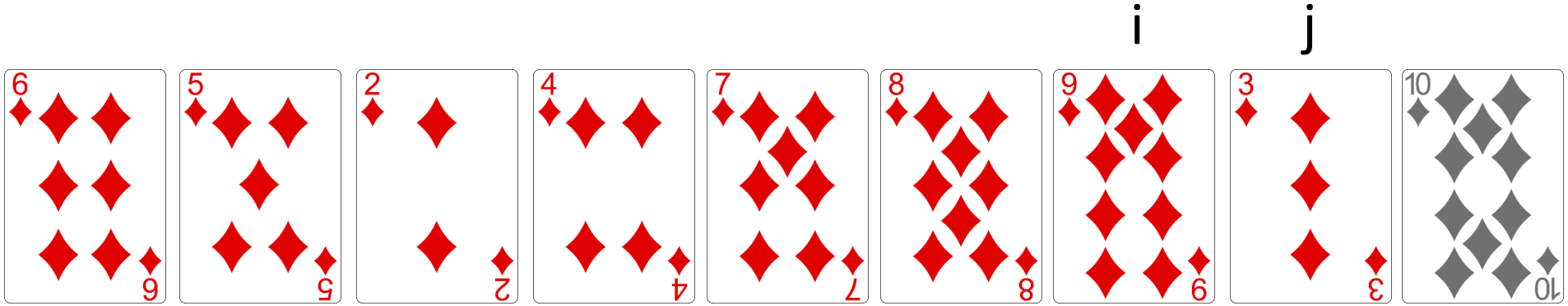

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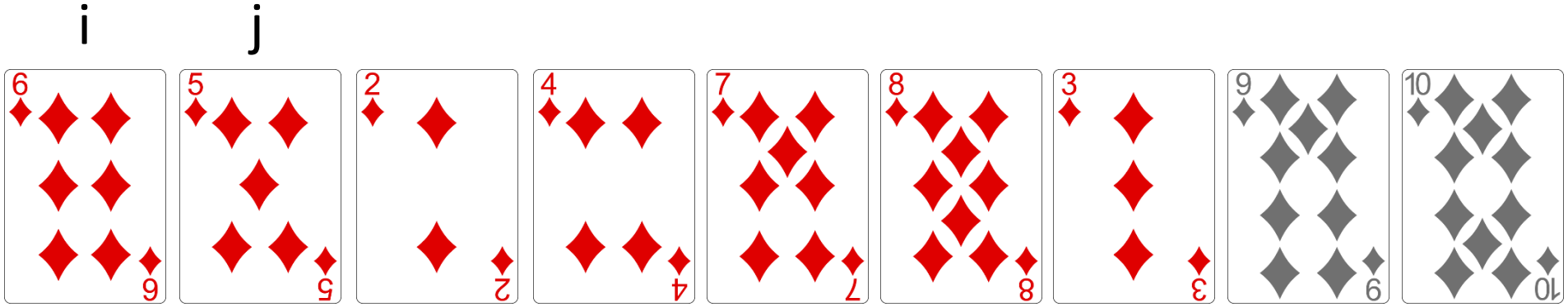

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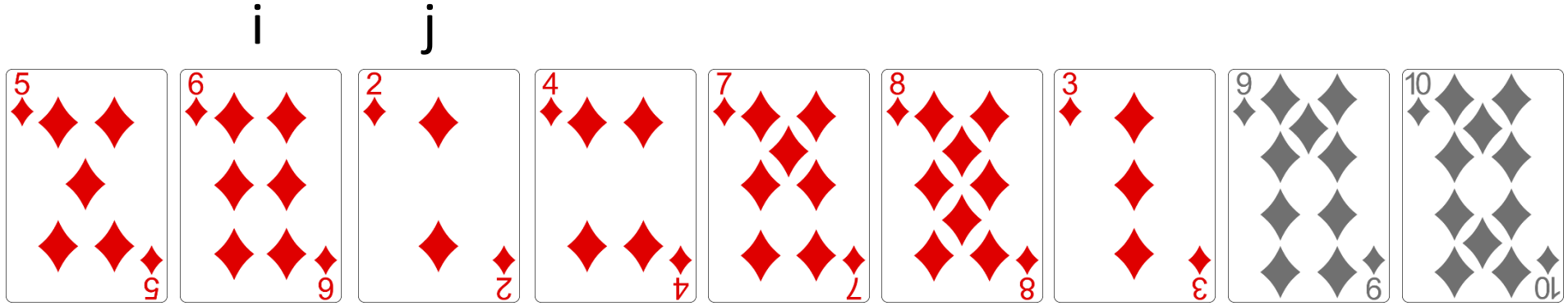

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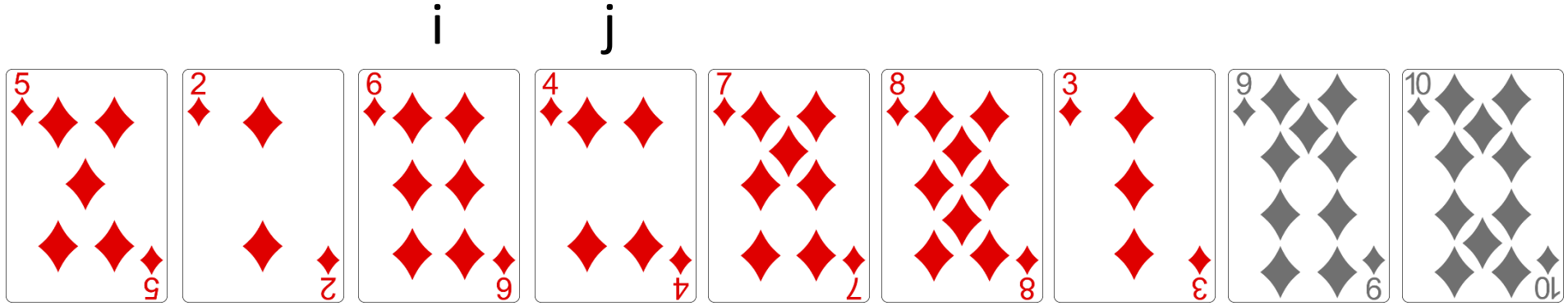

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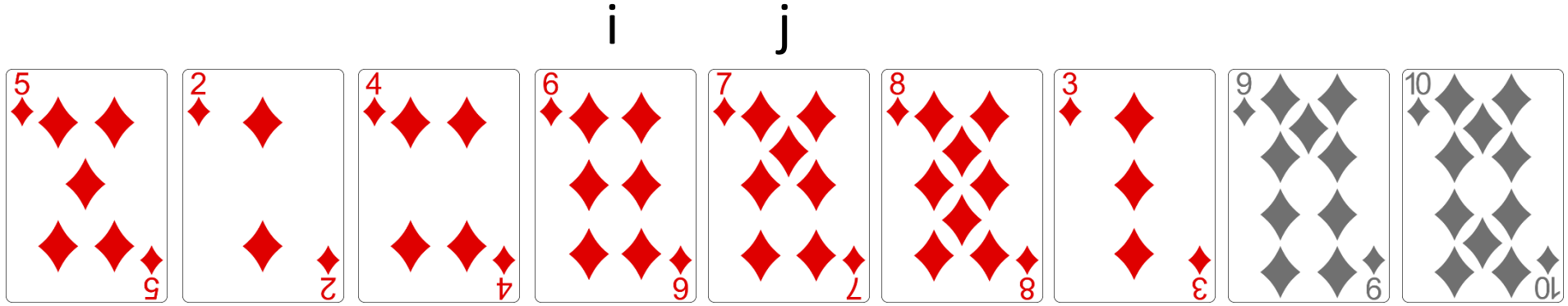

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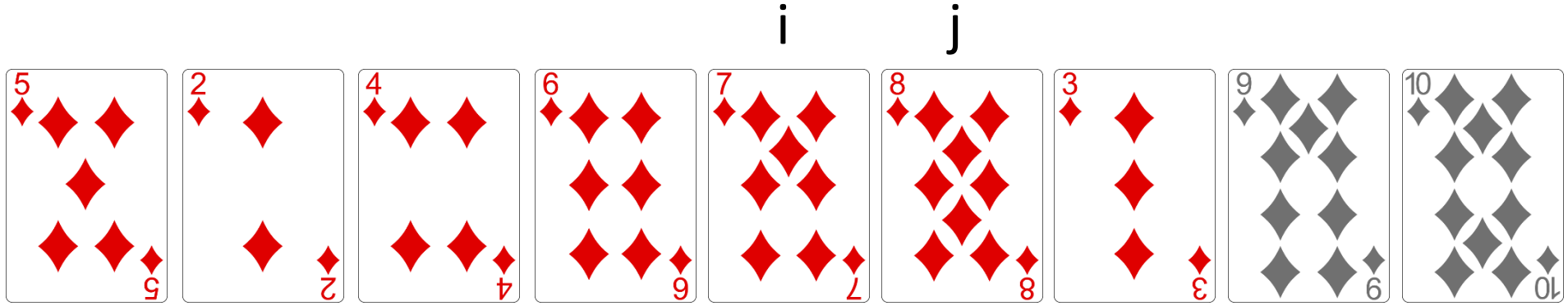

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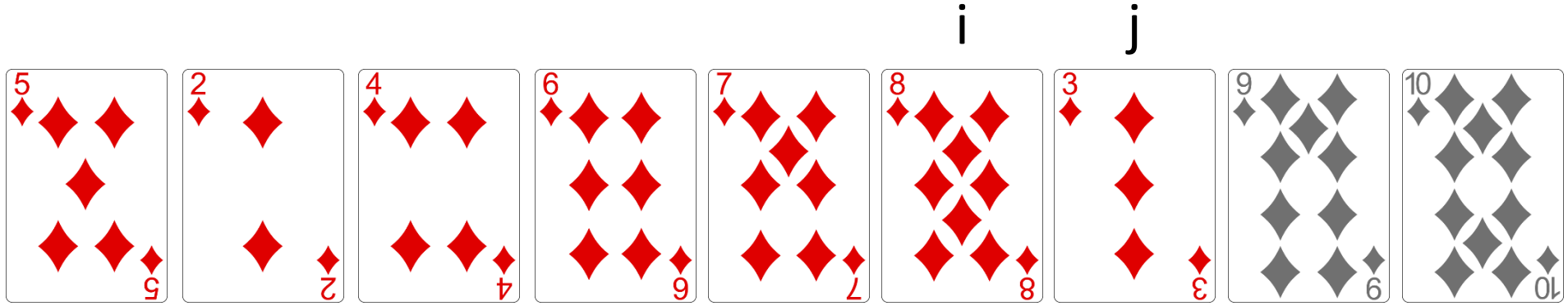

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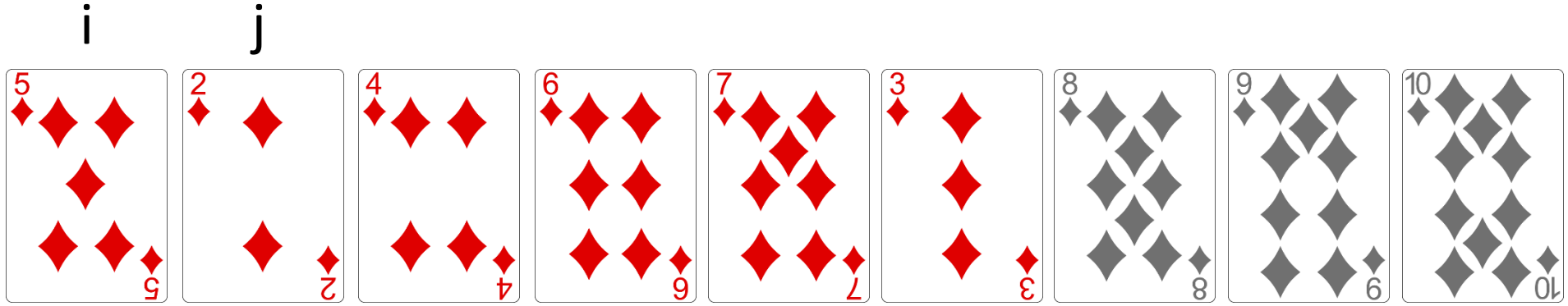

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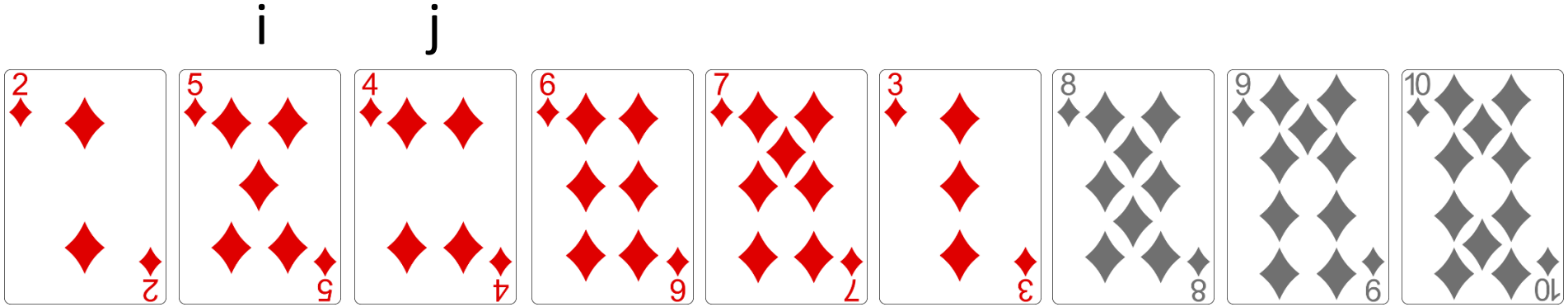

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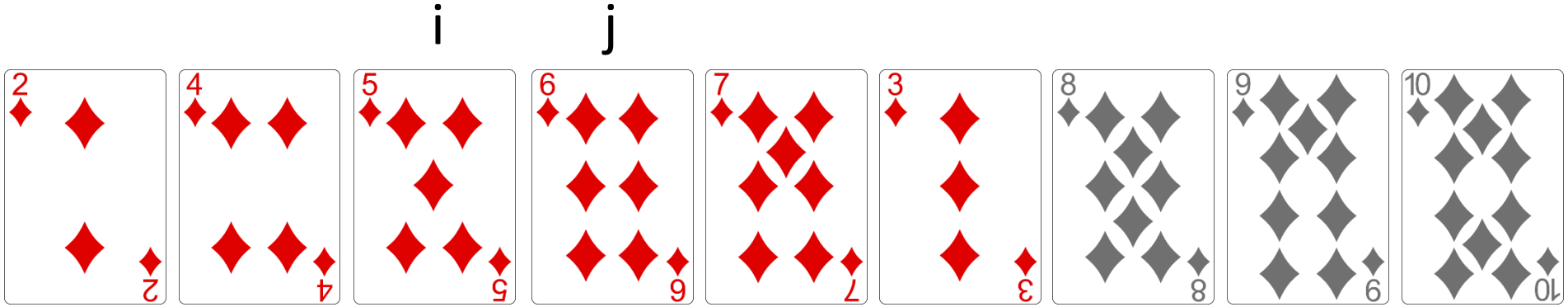

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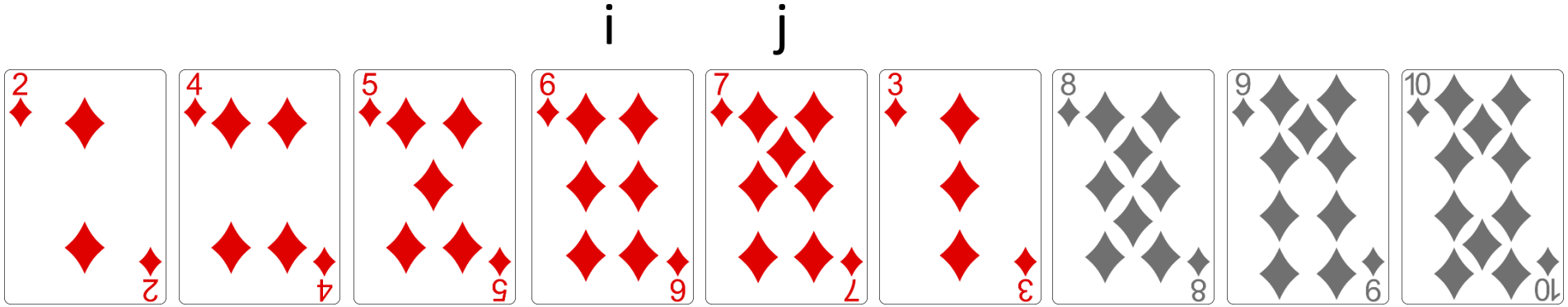

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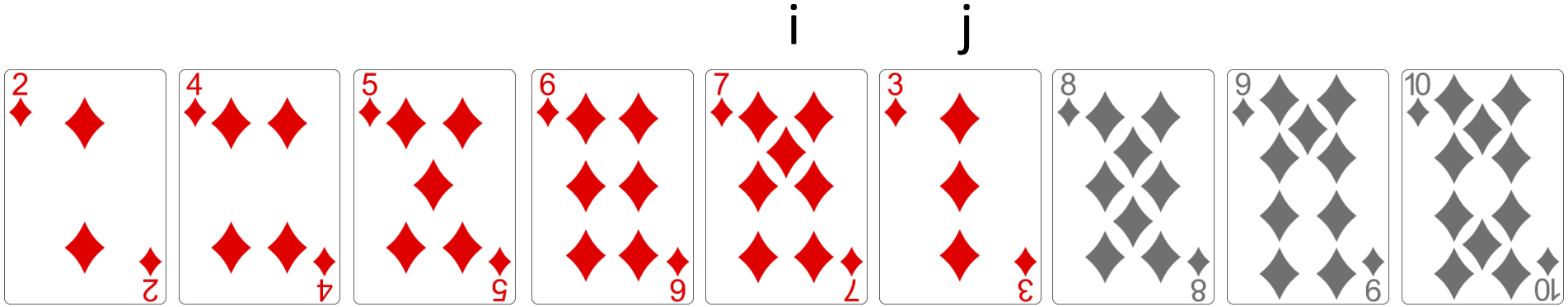

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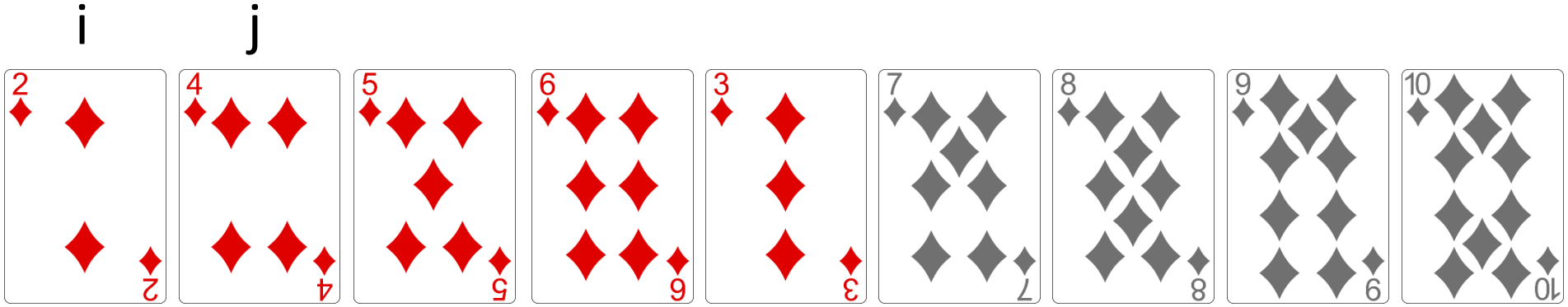

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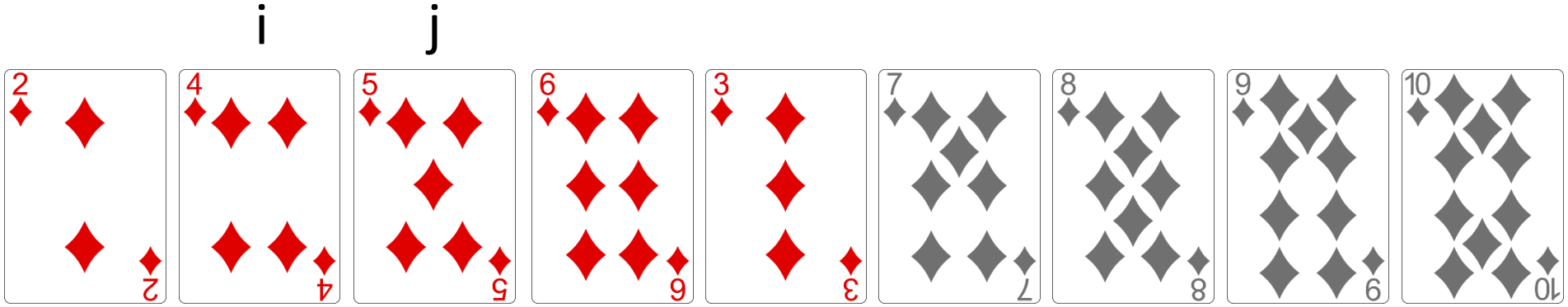

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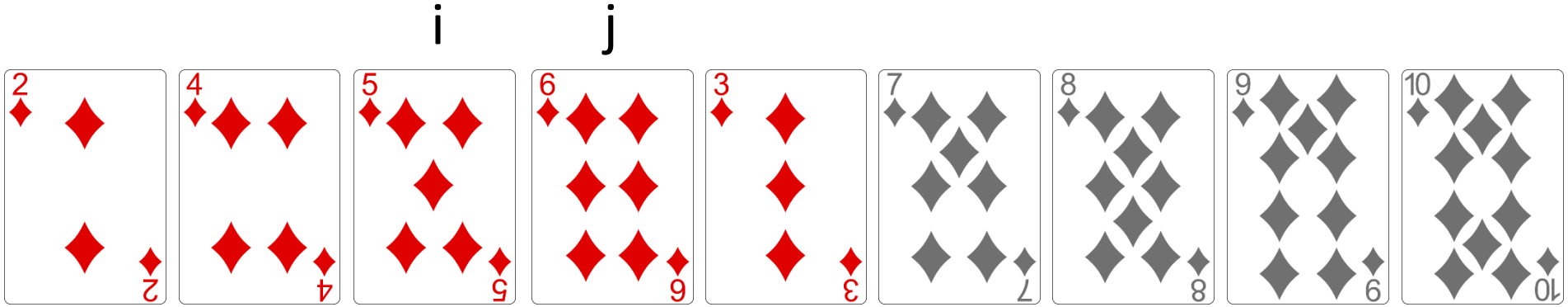

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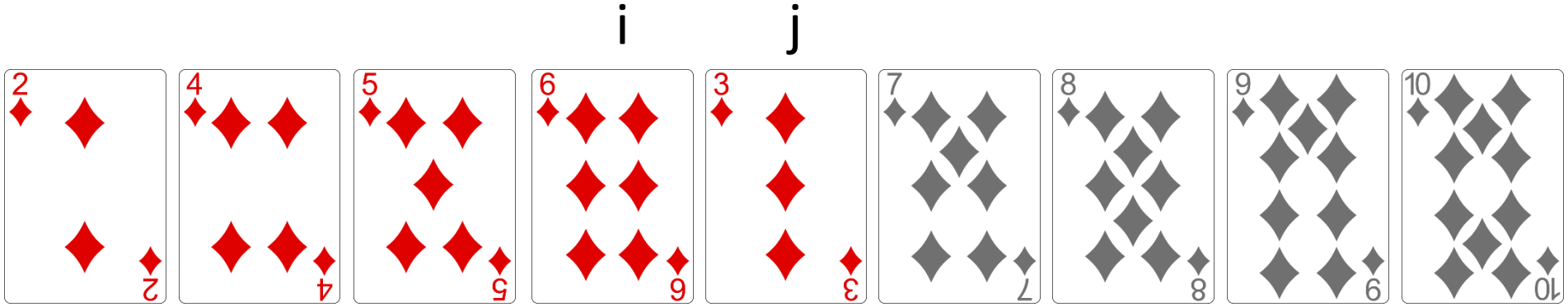

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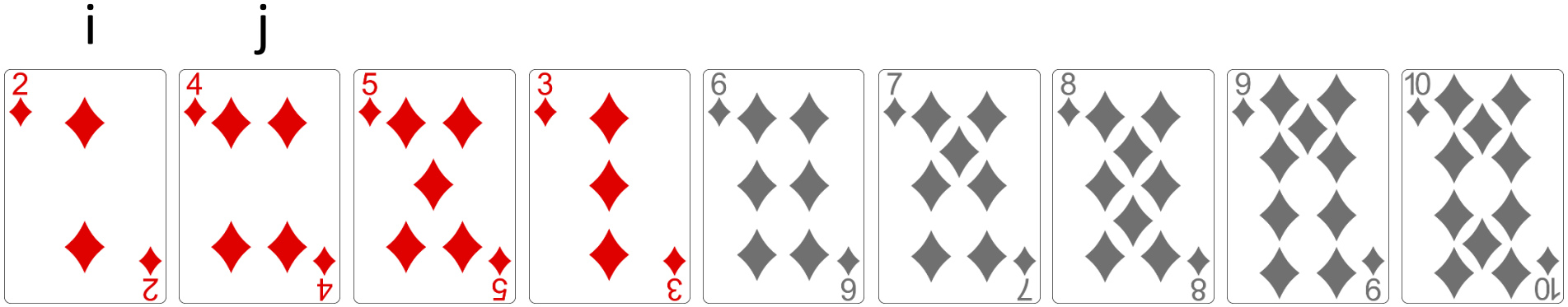

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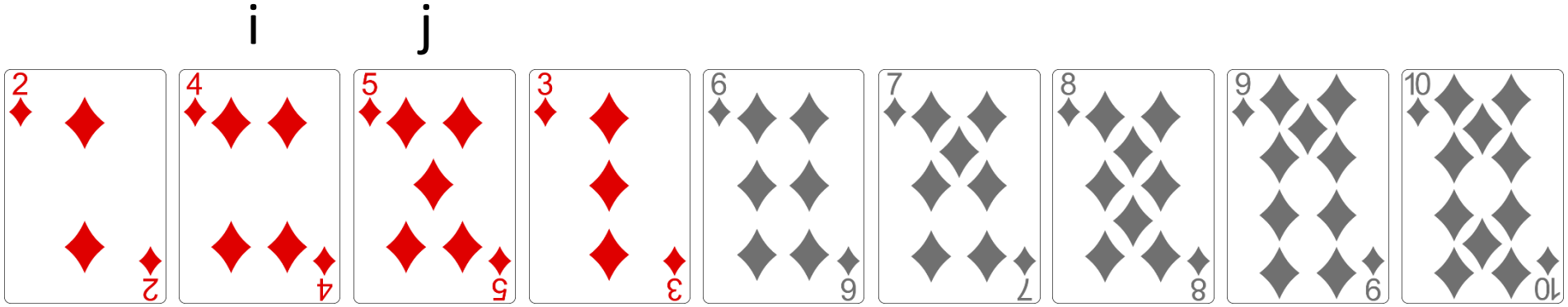

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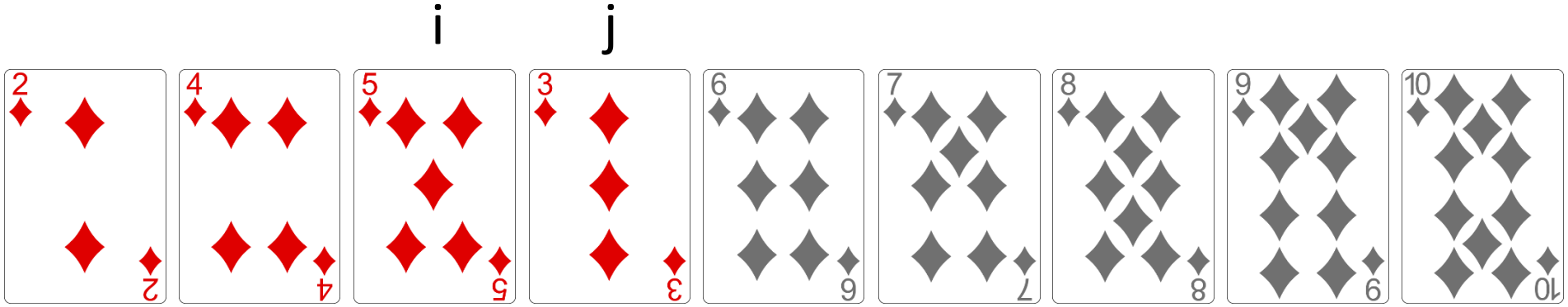

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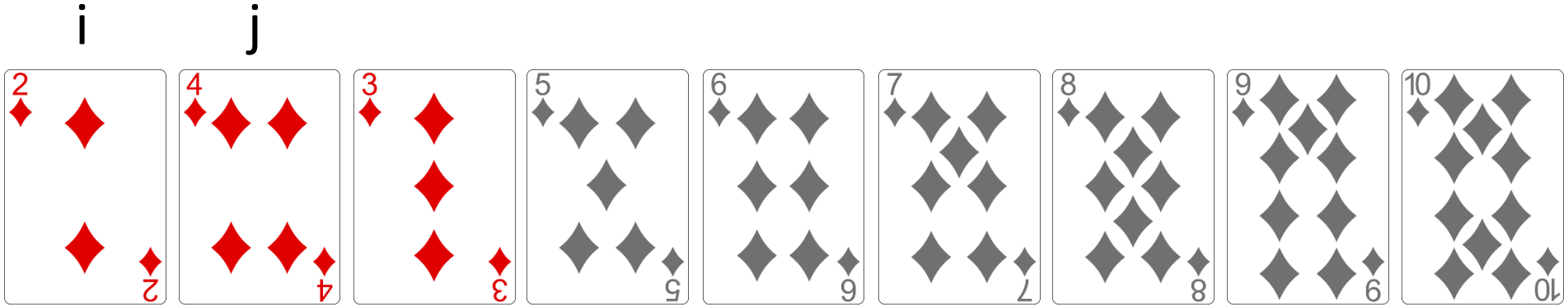

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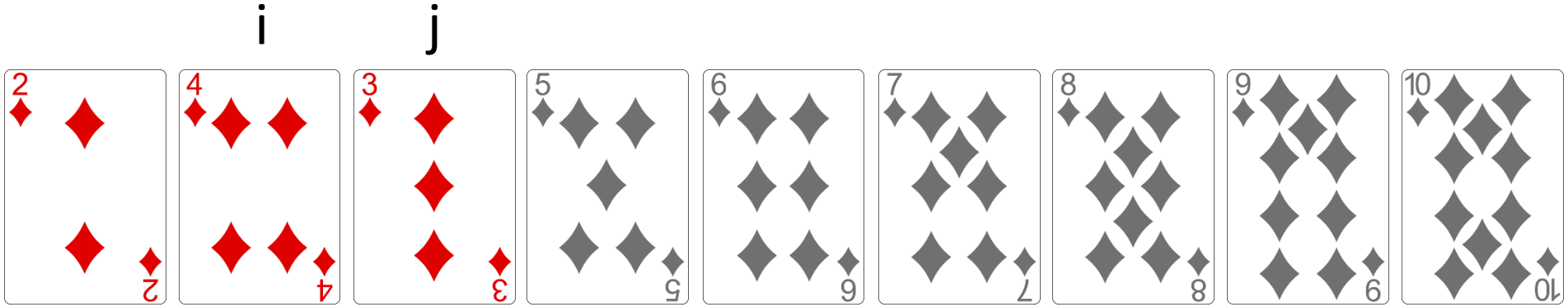

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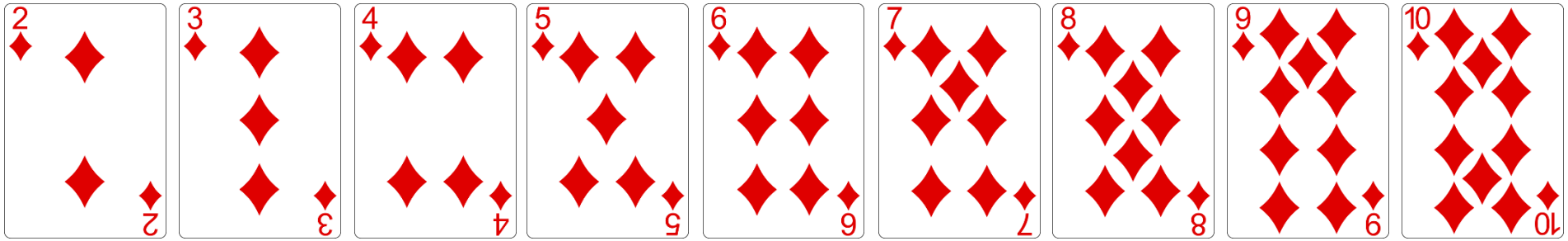


Bubble Sort Code

```
int[] numbers = {1, 3, 4, 2, 5, 6};
for (int i = 0; i < numbers.length; i++) {
  for (int j = i + 1; j < numbers.length - 1; j++) {
    if (numbers[i] > numbers[j]) {
      int tempNumber = numbers[i];
      numbers[i] = numbers[j];
      numbers[j] = tempNumber;
    }
  }
}
//TODO: Print numbers
```

Comparison of Sorting Algorithms

Name	Best	Average	Worst	Memory	Stable	Method
Selection	n^2	n^2	n^2	1	No	Selection
Bubble	n	n^2	n^2	1	Yes	Exchanging

Insertion Sort

- **Insertion Sort** – simple, but inefficient algorithm
  - Move the first unsorted element left to its place
  - Memory: O(1)
  - Stable: **Yes**
  - Method: **Insertion**

Insertion Sort

23 17 45 18 12 22

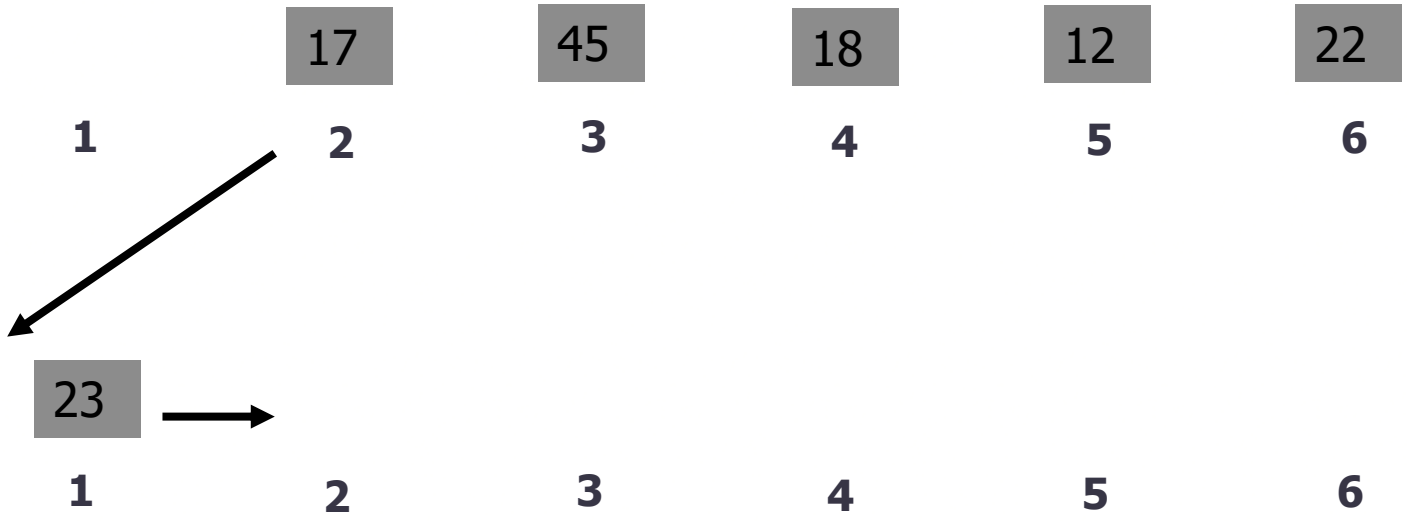
Insertion Sort

23	17	45	18	12	22
1	2	3	4	5	6

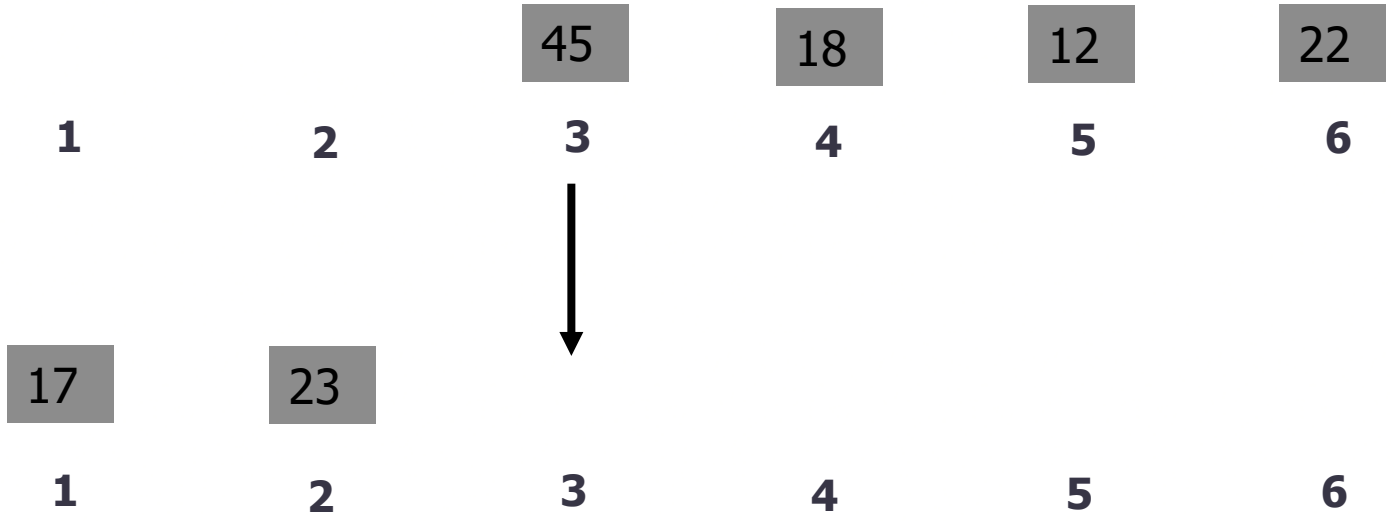
Insertion Sort

23	17	45	18	12	22
1	2	3	4	5	6
↓					
1	2	3	4	5	6

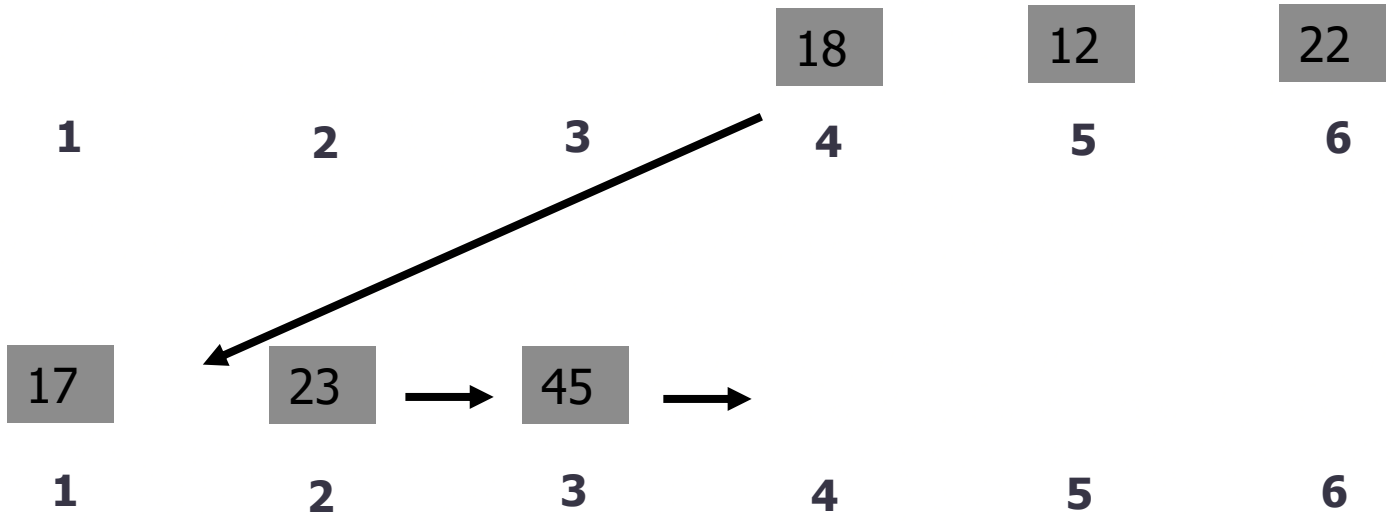
Insertion Sort



Insertion Sort



Insertion Sort



Insertion Sort

				12	22
1	2	3	4	5	6

17	18	23	45		
1	2	3	4	5	6

Insertion Sort

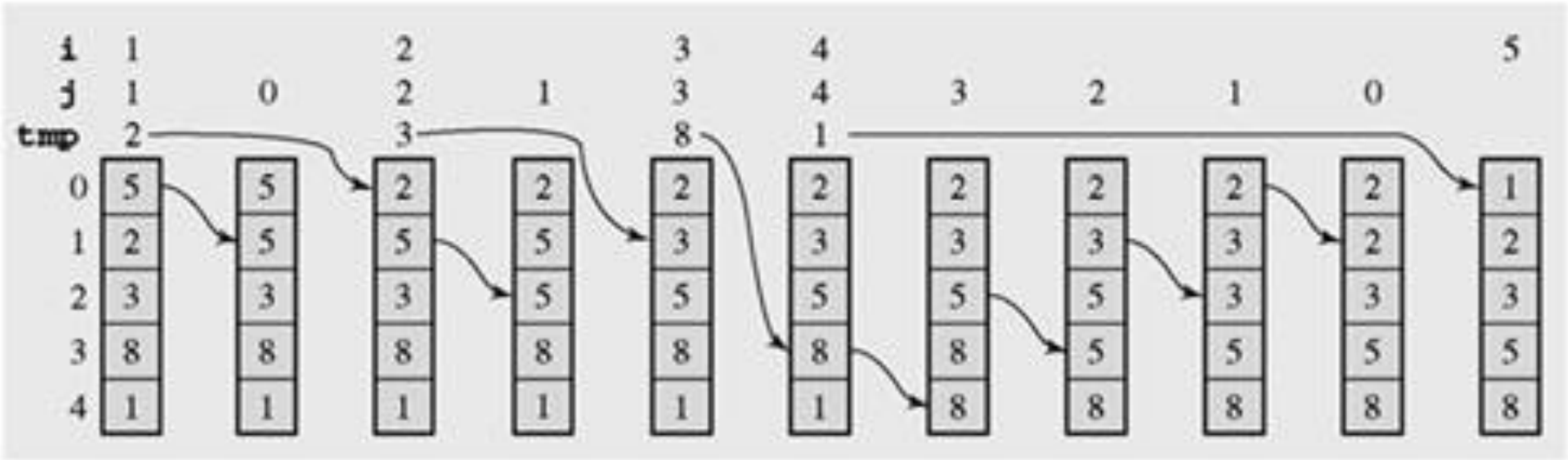
12	17	18	22	23	45
1	2	3	4	5	6

Insertion sort

- An insertion sort starts by considering the two first elements of the array data, which are data[0] and data[1]
- Next, the third element, data[2], is considered and inserted into its proper place (ascending)
- Its complexity is O(n2) and best case is O(n)

```
insertionsort(data[])
        for i=0 to data.length-1
                tmp=data[i];
                move all elements data[j] greater than tmp by one position;
                place tmp in its proper position;
```

Insertion sort example



The array [5 2 3 8 1] sorted by insertion sort

Insertion sort

```java
public static void main(String[] args) {
    int a[] = {5, 6, 1, 2, 4, 3, 7, 9, 8};
    insertionSort(a);
    for (int b : a) {
        System.out.print(b);
    }
    System.out.println("");
}
public static void insertionSort(int a[]){
    for (int i = 1; i < a.length; i++) {
        int temp = a[i];
        //find its correct place on the left.
        int j = i-1;
        while(j>=0 && a[j] > temp){
            //shift
            a[j+1] = a[j];
            j--;
        }
        a[++j] = temp;
    }
}
```

Comparison of Sorting Algorithms

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Bubble	n	n^2	n^2	1	Yes	Exchanging
Insertion	n	n^2	n^2	1	Yes	Insertion

Shuffling

- **Shuffling** == randomizing the order of items in a collection
  - Generate a random permutation
- *The generation of random numbers is too important to be left to chance.* —Robert R. Coveyou

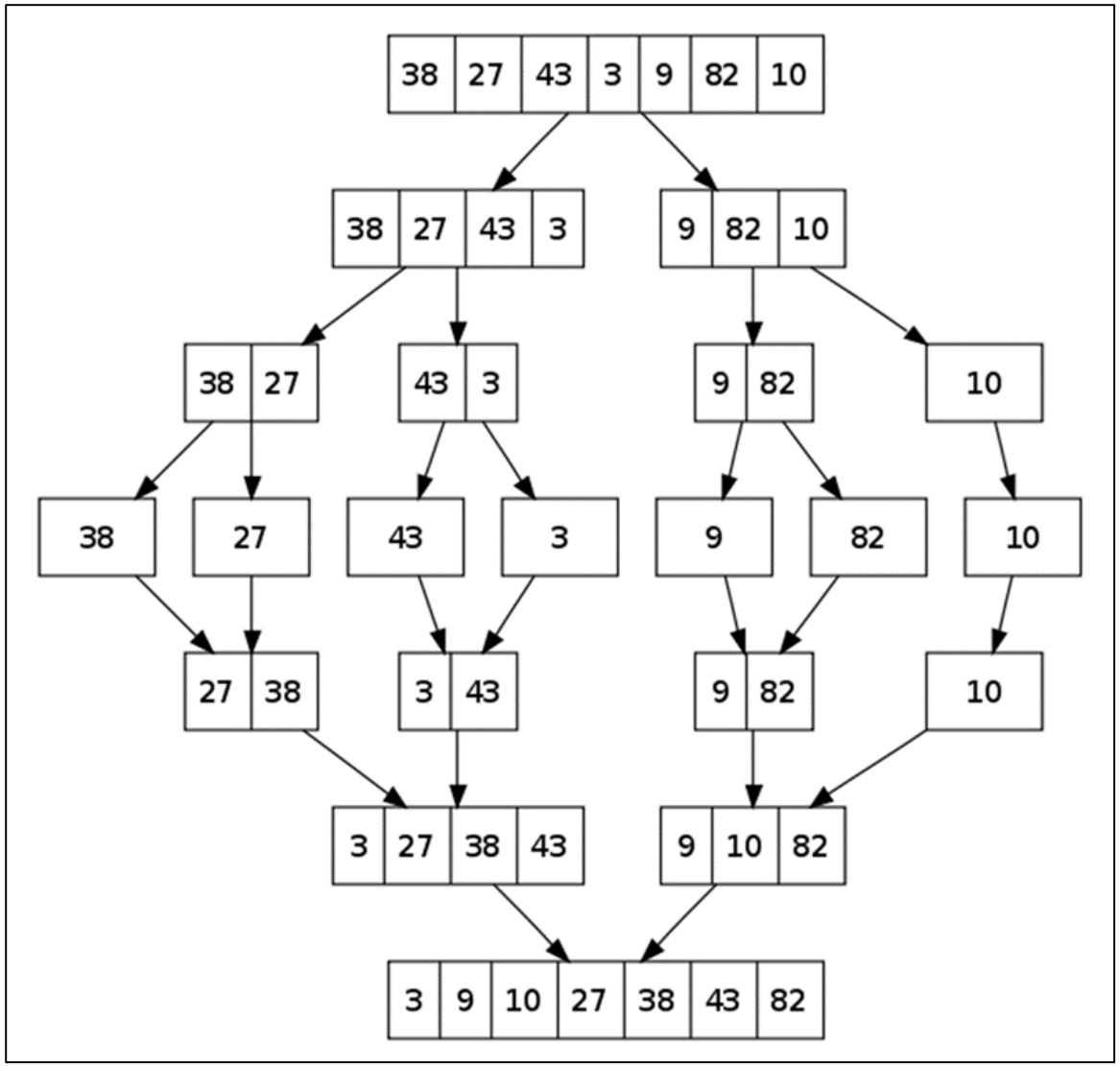
Fisher–Yates Shuffle Algorithm

```
public static <T> void shuffle(T[] source) {
    Random rand = new Random();
    for (int i = 0; i < source.length; i++) {
        // Exchange array[i] with random element in array[i … n-1]
        int r = i + rand.nextInt(source.length - i);
        T temp = source[i];
        source[i] = source[r];
        source[r] = temp;
    }
}
```

Merge Sort

- **Merge sort** is efficient sorting algorithm
- Divide the list into sub-lists (typically 2 sub-lists)
  1. Sort each sub-list (recursively call merge-sort)
  2. Merge the sorted sub-lists into a single list
- Best, average and worst case: **O(n*log(n))**
- Memory:
  - Typically
  - With in-place merge can be
- Highly parallelizable on multiple cores / machines → up to **O(log(n))**

Merge Sort: Hot It Works?



QuickSort

- **QuickSort** – efficient sorting algorithm
  - Choose a pivot; move smaller elements left & larger right; sort left & right
  - Best & average case: **O(n*log(n))**; Worst: **O(n2)**
  - Memory: **O(log(n))** stack space (for recursion)
  - Stable: **Depends**
  - Method: **Partitioning**

Counting Sort

- **Counting sort** is a very efficient sorting algorithm
- Sorts small integers by counting their occurrences
  - Not a comparison-based sort
  - Best, average and worst case: **O(n + k)**
    - **k** is the range of numbers to be sorted
    - E.g. [-1000 ... 1000] → **k** = 2001
- Memory: **O(n + k)** Space: **O(k)**
- Stable: Yes
- Method: Counting

Bucket Sort

- **Bucket sort** partitions an array into a number of buckets
  - Each bucket is then sorted individually with a different algorithm
  - Not a comparison-based sort
- Average case: **O(n + k)**
  - **k** = the number of buckets
- Worst case: **O(n * log n)**
- Stable: Yes (depends on algorithm)
- Memory: **O(n)** – **k** buckets holding n elements totally

Comparison of Sorting Algorithms

Name	Best	Average	Worst	Memory	Stable	Method
Selection	n^2	n^2	n^2	1	No	Selection
Bubble	n	n^2	n^2	1	Yes	Exchanging
Insertion	n	n^2	n^2	1	Yes	Insertion
Quick	n * log(n)	n * log(n)	n^2	1	Depends	Partitioning
Merge	n * log(n)	n * log(n)	n * log(n)	1 (or n)	Yes	Merging
Heap	n * log(n)	n * log(n)	n * log(n)	1	No	Selection
Bogo	n	n * n!	n * n!	1	No	Luck

Summary

- **Searching** algorithms
  - Binary Search, Interpolation Search
- **Slow** sorting algorithms:
  - Selection sort, Bubble sort, Insertion sort
- **Fast** sorting algorithms:
  - Quick sort, Merge sort, etc.
  - How to choose the most appropriate algorithm?
    - http://stackoverflow.com/a/1934004